

Analysis and Forecasting of Morocco's Outgoing International Mail Flows

Younes MLIK

Supervised by: EDDYA Moulay Lahcen

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1. Host Institution and Internship Context

1.1. Presentation of Barid Al-Maghrib

Barid Al-Maghrib (BAM) is the national postal operator of the Kingdom of Morocco. It is responsible for ensuring the provision of postal services across the entire national territory, within the framework of universal service obligations defined by Moroccan legislation. Over time, the institution has evolved from a traditional postal administration into a diversified group offering logistics, financial, and digital services.

1.1.1. Historical Background

The origins of postal services in Morocco date back to the end of the nineteenth century. The first structured postal organization was established by royal decree in 1892, marking the beginning of a formalized communication system within the country [1], [2].

During the Protectorate period, postal, telegraph, and telephone services were reorganized following European administrative models. In 1912, the Administration Chérifienne des Postes, des Télégraphes et des Téléphones was created and became responsible for managing national postal and telecommunications operations. The same year saw the issuance of the first Moroccan postage stamp, symbolizing the institutionalization of postal services [1].

After independence in 1956, postal and telecommunications services were integrated under the supervision of the Ministry of Posts, Telegraphs and Telephones. This period was characterized by efforts to extend postal coverage to rural areas and strengthen national infrastructure.

In 1984, the Office National des Postes et des Télécommunications (ONPT) was created, centralizing postal and telecommunications activities within a single public entity [3].

A major reform occurred with the promulgation of Law 24-96, which reorganized the communications sector in Morocco. This law led to the separation of postal and telecommunications activities and established Barid Al-Maghrib as an autonomous public institution with legal and financial independence [4].

In 2010, Barid Al-Maghrib was transformed into a state-owned limited company (société anonyme), reinforcing its managerial autonomy while maintaining full public ownership. This transformation was accompanied by the consolidation of financial activities under Al Barid Bank [2].

1.1.2. Missions of Barid Al-Maghrib

Barid Al-Maghrib is entrusted with ensuring universal postal service coverage throughout Morocco. Its missions are defined by Law 24-96 and include the collection, sorting, transportation, and distribution of postal items at both national and international levels [4].

More specifically, BAM is responsible for:

- Providing universal postal services across all regions;
- Managing domestic and international mail and parcel flows;
- Operating express mail and logistics services;
- Contributing to financial inclusion through postal financial services;

- Supporting e-government and digital transformation initiatives;
- Ensuring compliance with international postal agreements.

These missions place BAM at the intersection of public service obligations and competitive logistics markets, requiring a balance between accessibility, affordability, and operational efficiency.

1.1.3. Organizational Structure

Barid Al-Maghrib is structured as a hierarchical organization governed by a Board of Directors responsible for defining strategic orientations and approving major investment and development plans.

Operational management is ensured by executive directorates covering core functions such as:

- Postal operations and logistics;
- Information systems and digital transformation;
- Finance and accounting;
- Human resources management;
- Commercial and customer services.

The operational network is distributed across regional directorates, sorting centers, and post offices, ensuring nationwide coverage. This decentralized structure enables the institution to maintain proximity to users while coordinating large-scale logistics operations.

The governance framework of BAM is defined by Law 24-96 and subsequent regulatory texts governing public enterprises in Morocco [4].

1.1.4. Postal Activities

Historically, postal services constituted the core activity of Barid Al-Maghrib. While traditional letter mail has declined due to digital communication, parcel and logistics activities have experienced strong growth, driven mainly by the expansion of e-commerce.

The main postal products include:

- Ordinary mail and printed matter;
- Registered mail with tracking capabilities;
- Express Mail Service (EMS);
- Domestic and international parcels;
- Post office boxes and address-related services.

International parcel flows in particular have become a strategic activity due to increasing trade exchanges and diaspora-related shipments. These services require strong coordination between operational, customs, and transportation systems.

1.1.5. Subsidiaries and Group Structure

Barid Al-Maghrib has developed a group structure to diversify its activities and respond to evolving market demands.

The most important subsidiary is Al Barid Bank, created in 2010 to consolidate postal financial services and promote financial inclusion in Morocco. Leveraging BAM's extensive territorial

network, Al Barid Bank provides banking services to individuals and small businesses across the country [2].

Other affiliated entities operate in logistics, express delivery, and media services, contributing to the diversification of the group's revenue sources and strengthening its position in competitive markets.

1.1.6. International Cooperation

Barid Al-Maghrib is a member of the Universal Postal Union (UPU), a specialized agency of the United Nations responsible for coordinating international postal exchanges and establishing global postal standards [5].

Through this membership, Morocco participates in a global postal network that enables the exchange of mail with more than 180 countries. The UPU framework defines operational rules governing:

- International dispatch preparation;
- Customs documentation;
- Tracking and electronic data exchange;
- Compensation between postal operators.

Compliance with these standards ensures interoperability between national postal systems and improves the reliability of cross-border postal flows.

1.1.7. Challenges and Strategic Orientation

Like many postal operators worldwide, Barid Al-Maghrib faces structural challenges related to the decline of traditional mail volumes and the rapid growth of parcel logistics driven by e-commerce.

Key challenges include:

- Increasing variability in international shipment volumes;
- Modernization of legacy information systems;
- Optimization of transportation and sorting capacities;
- Improvement of shipment traceability and data quality;
- Integration of data-driven decision-making tools.

To address these challenges, BAM has engaged in a strategic transformation process focused on digitalization, operational efficiency, and service diversification. The exploitation of data and predictive analytics has become increasingly important in supporting planning and decision-making processes.

In this context, the present internship contributes to the broader objective of enhancing operational intelligence through the analysis and forecasting of Morocco's outgoing international mail flows.

1.2. International Mail Activities

International mail activities constitute a strategic component of Barid Al-Maghrib's operations. They enable the exchange of letters, documents, and parcels between Morocco

and foreign destinations through a globally standardized postal network. These operations are governed by international conventions, customs regulations, and operational standards defined mainly by the Universal Postal Union (UPU) [5].

The continuous expansion of global trade, the growth of e-commerce, and the importance of the Moroccan diaspora have significantly increased the volume and complexity of international postal flows. As a result, international mail has evolved from a traditional correspondence service into a critical logistics and cross-border commerce infrastructure.

1.2.1. Universal Postal Framework

The international postal system is structured around the Universal Postal Union (UPU), a specialized agency of the United Nations founded in 1874. Its primary objective is to coordinate postal policies among member countries and ensure a unified international postal territory for the exchange of mail [6].

Under this framework, postal operators such as Barid Al-Maghrib are designated operators responsible for:

- Ensuring exchange of international mail;
- Applying standardized postal tariffs and formats;
- Guaranteeing interoperability of tracking systems;
- Managing compensation mechanisms between countries;
- Ensuring compliance with customs and security regulations.

The UPU establishes a set of technical standards known as the Postal Operations Manual, which defines how international mail must be processed, routed, and tracked across postal networks [7].

These standards are essential to ensure end-to-end interoperability between heterogeneous national postal systems.

1.2.2. Structure of International Postal Exchanges

International mail exchanges are based on bilateral and multilateral agreements between designated postal operators. Morocco exchanges mail with more than 180 countries through a structured logistics network composed of air transport carriers, international exchange offices, and foreign postal administrations [8].

The international flow is generally divided into two main categories:

- Outgoing international mail: shipments originating from Morocco and destined abroad;
- Incoming international mail: shipments arriving in Morocco from foreign countries.

This study focuses exclusively on outgoing international mail flows.

The routing of international mail depends on:

- Destination country;
- Transport availability;
- Customs requirements;
- Service level (standard, express, registered).

1.2.3. International Postal Products

Barid Al-Maghrib offers several categories of international postal products adapted to different customer needs in terms of speed, reliability, and cost.

1.2.3.1. Ordinary International Mail

Ordinary mail includes letters, documents, and low-value parcels. These items are transported through the international postal network without enhanced tracking guarantees.

Although volume has decreased due to digital communication, this category remains relevant for administrative and legal correspondence.

1.2.3.2. Registered Mail

Registered international mail provides a tracking number and proof of deposit. It ensures greater security and is commonly used for legal documents and official correspondence.

Registered items benefit from:

- Unique identification codes;
- Tracking visibility at major transit points;
- Delivery confirmation in destination country.

1.2.3.3. Express Mail Service (EMS)

The Express Mail Service (EMS) is a premium international service developed under the coordination of the UPU. It provides:

- Priority handling;
- Faster transit times;
- End-to-end tracking;
- Delivery confirmation.

EMS is particularly used for urgent documents and high-value shipments [9].

1.2.3.4. International Parcels

International parcels represent the fastest-growing segment due to e-commerce expansion. These shipments include:

- Consumer goods;
- Gifts;
- Commercial products;
- Personal belongings.

They may include optional services such as insurance, declared value, or proof of delivery depending on destination constraints.

1.2.4. Operational Flow of Outgoing International Mail

The operational processing of outgoing international mail follows a multi-stage logistics pipeline.

1.2.4.1. Acceptance Stage

Shipments are first deposited at post offices or authorized collection centers. At this stage, the following information is recorded:

- Sender and recipient details;
- Destination country;
- Weight and dimensions;
- Service type;
- Customs declaration (if applicable).

Each shipment is assigned a unique identifier enabling tracking throughout the postal network.

1.2.4.2. Sorting and Consolidation

After acceptance, items are transferred to sorting centers where they are grouped according to destination and transport route.

Operations include:

- Barcode scanning;
- Automated or manual sorting;
- Dispatch creation;
- Containerization into postal sacks.

Modern sorting systems rely heavily on information systems to ensure accuracy and traceability.

1.2.4.3. Customs Processing

International shipments containing goods are subject to customs controls. This step involves:

- Verification of declared content;
- Validation of value and classification;
- Compliance with import/export regulations.

Customs procedures represent one of the main sources of delay in international postal logistics.

1.2.4.4. Dispatch and Transportation

Sorted mail is consolidated into dispatch units and transported via air or land carriers to destination countries or transit hubs.

Air transport remains the dominant mode for international postal flows due to speed and global coverage.

1.2.4.5. Exchange with Foreign Operators

Upon arrival in the destination country, mail is transferred to the designated postal operator, which ensures final delivery.

End-to-end tracking visibility depends on data exchange between postal operators and the interoperability of their information systems.

1.2.5. Characteristics of Moroccan Outgoing International Mail

Outgoing international mail from Morocco exhibits several structural characteristics.

First, a large proportion of shipments originate from major urban centers such as Casablanca, Rabat, Marrakech, Tangier, and Agadir. These cities concentrate economic activity, diaspora-related services, and e-commerce transactions [10].

Second, destination countries are highly diversified, reflecting Morocco's global commercial and migratory links. Europe remains the primary destination region, followed by North America and the Middle East.

Third, the flow exhibits strong temporal seasonality. Empirical analysis conducted during this internship shows a significant increase in shipment volumes prior to Ramadan periods. This pattern is consistent with cultural and religious practices involving gift exchange and remittances among diaspora communities.

Other influencing factors include:

- E-commerce promotional cycles;
- Summer holiday periods;
- International shipping disruptions;
- Exchange rate fluctuations affecting cross-border trade.

1.2.6. Operational Challenges

International postal operations face multiple structural and operational challenges.

One of the main challenges is demand variability. Shipment volumes fluctuate significantly over time, complicating capacity planning and resource allocation.

Another challenge is customs dependency. Clearance times vary depending on destination country regulations and the nature of goods transported.

Additionally, interoperability between postal operators remains uneven. While the UPU provides standardized frameworks, implementation varies across countries.

Finally, legacy information systems often prioritize operational processing rather than analytical exploitation, making large-scale data extraction difficult for forecasting and decision-support applications.

These constraints justify the need for advanced data-driven approaches to improve forecasting accuracy and operational planning.

1.2.7. Strategic Importance

International mail activities are increasingly strategic for Barid Al-Maghrib due to the rise of cross-border e-commerce and diaspora-driven demand.

They contribute to:

- Foreign exchange inflows;
- Economic integration with global markets;
- Service diversification of the postal operator;
- Strengthening Morocco's logistics infrastructure.

In this context, improving the predictability of outgoing international mail flows is essential for optimizing transport planning, resource allocation, and service quality.

1.3. Information Systems

The information systems of Barid Al-Maghrib (BAM) constitute the technological backbone supporting postal operations, logistics management, and customer services. These systems are responsible for managing the end-to-end lifecycle of postal items, from acceptance at post offices to final delivery or international exchange.

Over the last decades, BAM has progressively modernized its information systems to accompany the growth of parcel logistics, the diversification of services, and the increasing demand for real-time tracking and operational visibility.

Despite these advancements, several components of the system remain legacy-oriented and primarily designed for operational execution rather than analytical exploitation. This characteristic has a direct impact on data accessibility and the ability to perform large-scale data-driven analysis.

1.3.1. Digital Transformation Context

Like many historical postal operators, Barid Al-Maghrib is engaged in a digital transformation process aimed at improving operational efficiency, enhancing customer experience, and enabling data-driven decision-making.

This transformation is driven by several factors:

- Growth of e-commerce and parcel logistics;
- Increasing demand for shipment traceability;
- Need for real-time operational monitoring;
- Expansion of digital financial services;
- Pressure from private logistics competitors.

According to international postal modernization frameworks, digital transformation in postal operators typically involves the integration of advanced information systems, automation of sorting processes, and adoption of analytics platforms for operational optimization [11].

Within this context, BAM operates multiple internal systems that support different functional domains of postal operations.

1.3.2. Système de Messagerie Intégré (SMI)

The Système de Messagerie Intégré (SMI) is the core operational information system used by Barid Al-Maghrib to manage parcel and mail activities.

SMI is primarily designed to:

- Record shipment acceptance data;

- Track operational events;
- Manage sorting and routing operations;
- Provide operational reporting interfaces;
- Support customer service tracking queries.

SMI plays a central role in the lifecycle of postal items, as it stores all key events associated with a shipment, including deposit, transit, sorting, dispatch, and delivery status updates.

However, SMI is fundamentally an operational system (OLTP-oriented) and not designed for large-scale analytical workloads. As a result, data extraction for research or forecasting purposes presents several constraints:

- Absence of a documented public API;
- Limited bulk export capabilities;
- Performance degradation for large queries;
- Interface designed for manual consultation;
- Lack of structured historical datasets for analytics.

These limitations significantly influenced the design of the data acquisition strategy used in this internship.

1.3.3. Tracking and Event Management Systems

Tracking systems within BAM provide visibility over the status of shipments throughout their lifecycle.

Each shipment is associated with a unique identifier that allows retrieval of operational events such as:

- Acceptance at post office;
- Sorting center processing;
- Dispatch to destination hub;
- Arrival at destination;
- Delivery or return status.

Tracking data is generated as a sequence of timestamped events stored in operational logs.

While the tracking system enables customer-facing visibility, its backend is optimized for individual queries rather than batch processing. This makes large-scale extraction of historical tracking data technically challenging.

In this study, tracking information was retrieved indirectly through automated interaction with the web interface due to the absence of bulk export mechanisms.

1.3.4. Reporting and Data Extraction Layer

SMI provides a set of reporting interfaces used internally for operational monitoring and management decision-making.

These reporting modules allow users to generate aggregated views of:

- Daily shipment volumes;
- Destination breakdowns;

- Agency-level performance;
- Delivery status distributions;
- Service-level statistics.

However, these reporting tools exhibit several structural limitations:

- Query execution time increases significantly for long time ranges;
- Some reports become unstable beyond six months of data;
- Outputs are primarily designed for visualization rather than structured data export;
- No standardized API exists for automated extraction.

As a result, data extraction for analytical purposes requires alternative methods such as browser automation and scraping-based approaches.

This constraint directly motivated the development of a custom Robotic Process Automation (RPA) pipeline using Playwright for Python [12].

1.3.5. Data Architecture and Operational Flow

The operational data architecture of BAM can be conceptually described as a multi-layer system composed of:

- Operational layer (SMI);
- Tracking layer (event logs);
- Reporting layer (web interfaces);
- External exchange systems (international postal partners).

At the core, SMI acts as the system of record for all postal events. Each operational action performed on a parcel generates a corresponding event entry, which is stored and later aggregated for reporting purposes.

The absence of a unified analytical data warehouse means that historical data reconstruction often requires combining multiple sources, including:

- Web reports;
- Tracking HTML pages;
- Exported CSV files;
- Operational logs.

This fragmented architecture increases the complexity of data engineering tasks and necessitates custom ETL pipelines for analytical use cases.

1.3.6. Data Engineering Constraints

During this internship, several technical constraints were identified in the data acquisition process:

- Lack of API access for historical data;
- Rate limiting and system instability for large queries;
- HTML-based tracking pages not designed for structured extraction;
- Session-based authentication requiring automation;
- Inconsistent export formats across modules.

To overcome these constraints, a robust extraction pipeline was implemented using browser automation, retry mechanisms, and incremental data collection strategies.

This pipeline ensured:

- Fault tolerance in case of session expiration;
- Resume capability after interruptions;
- Idempotent data collection;
- Efficient storage of intermediate results.

The final dataset was stored in compressed columnar formats (Parquet) to optimize analytical performance.

1.3.7. Role of Information Systems in This Study

The information systems described above play a central role in this internship, as they constitute the primary source of raw operational data.

The forecasting task developed in this study relies entirely on historical data extracted from SMI and associated tracking systems.

In particular:

- SMI provides shipment-level operational events;
- Tracking systems provide lifecycle state transitions;
- Reporting systems provide aggregated validation metrics.

The combination of these sources enables the reconstruction of historical international mail flows, which are then used for time-series modeling and forecasting.

This dependency highlights the importance of understanding legacy information systems when designing data-driven analytical solutions in operational environments.

1.3.8. Summary

Barid Al-Maghrib's information systems form a complex ecosystem combining operational processing systems, tracking infrastructures, and reporting tools.

While these systems ensure reliable day-to-day postal operations, their architecture is not fully optimized for analytical exploitation. This creates significant challenges for large-scale data extraction and forecasting applications.

The methodological approach adopted in this internship was therefore designed to bridge the gap between operational systems and analytical requirements through automation, data engineering, and structured preprocessing pipelines.

1.4. Internship Context

This internship was carried out within Barid Al-Maghrib (BAM), the national postal operator of Morocco, as part of the requirements for the completion of an engineering curriculum. The objective of the internship is to apply theoretical knowledge in a real industrial environment while contributing to a concrete data-driven project related to operational forecasting.

The internship focuses on the analysis and forecasting of outgoing international mail flows, a strategic operational domain within BAM. This context provides both a technical and organizational framework in which large-scale postal data is collected, processed, and exploited for decision-support purposes.

1.4.1. Industrial Context of the Internship

Barid Al-Maghrib is a state-owned enterprise responsible for postal, logistics, financial, and digital services across the Moroccan territory [13].

The organization operates a large national network covering urban and rural areas and manages millions of postal transactions annually. Its activities include:

- Domestic and international mail services;
- Parcel and express logistics;
- Financial services through its subsidiary Al Barid Bank;
- Digital trust services (e-signature, certification);
- E-commerce logistics support.

Within this industrial context, international mail operations represent a critical segment due to their direct link with global trade flows and diaspora-related demand.

1.4.2. Internship Department and Environment

The internship was conducted in an operationally oriented environment closely related to postal data management and performance analysis.

The working environment is characterized by:

- Strong reliance on legacy operational information systems;
- Fragmented data sources distributed across multiple platforms;
- Limited availability of analytical interfaces;
- High dependence on operational reporting tools;
- Continuous production constraints requiring robust systems.

The internship work was primarily focused on extracting and structuring data from operational systems in order to build a reliable dataset suitable for forecasting models.

1.4.3. Internship Mission

The main mission of the internship can be summarized as follows:

- Analyze historical outgoing international mail flows;
- Build a structured dataset from heterogeneous operational sources;
- Design and implement an automated data extraction pipeline;
- Perform exploratory data analysis to identify temporal patterns;
- Develop forecasting models for operational planning;
- Evaluate model performance using statistical metrics.

This mission is aligned with the strategic objective of improving decision-making processes within BAM through data-driven approaches.

1.4.4. Tools and Technologies Used

The internship required the use of several tools and technologies adapted to both data engineering and statistical analysis tasks.

The main tools used include:

- Python: core language for data processing and modeling;
- Playwright: browser automation for data extraction [12];
- Pandas: data manipulation and transformation;
- Apache Parquet: optimized columnar storage format for large datasets;
- Statistical libraries: forecasting and time series modeling tools;
- Jupyter-style workflows for iterative analysis.

These tools were selected to address the lack of structured APIs and the need for scalable data extraction from operational systems.

1.4.5. Data Context and Availability

The data used in this internship originates from multiple internal operational systems, primarily the Système de Messagerie Intégré (SMI).

These systems provide information such as:

- Shipment deposit dates;
- Origin and destination details;
- Weight and dimensional attributes;
- Operational status updates;
- Delivery and routing information.

However, the data environment presents several constraints:

- Absence of a dedicated analytical data warehouse;
- Limited bulk export capabilities;
- Reliance on HTML-based reporting interfaces;
- Inconsistent historical coverage in some modules;
- System performance limitations for large queries.

These constraints required the design of a robust extraction and preprocessing pipeline capable of reconstructing consistent historical datasets.

1.4.6. Position of the Internship within BAM Activities

This internship is positioned at the intersection of operational postal activities and emerging data analytics initiatives.

Traditionally, BAM systems are optimized for operational execution rather than predictive analytics. As a result, forecasting tasks are not natively integrated into the core systems.

The internship therefore contributes to:

- Bridging operational systems and analytical modeling;
- Transforming raw operational data into structured datasets;

- Supporting capacity planning and logistics optimization;
- Enhancing visibility on international mail flows.

1.4.7. Expected Contribution

The expected contribution of this internship is twofold.

First, from a technical perspective, it provides a complete data pipeline capable of extracting, cleaning, and structuring large-scale postal datasets.

Second, from an analytical perspective, it develops forecasting models capable of estimating future outgoing international mail volumes over a defined horizon.

This contributes to:

- Improved operational planning;
- Better resource allocation;
- Enhanced understanding of temporal demand patterns;
- Support for digital transformation initiatives within BAM.

1.4.8. Summary

The internship is conducted within a complex industrial and information system environment where operational constraints strongly influence data availability and usability.

Despite these limitations, the internship provides a structured framework for applying data engineering and time series forecasting techniques to real postal operations. It serves as a practical bridge between academic knowledge and industrial requirements.

1.5. Problem Statement

The objective of this internship is to address a practical operational problem faced within Barid Al-Maghrib: the lack of reliable forecasting tools for outgoing international mail flows.

Despite the existence of operational information systems such as the *Système de Messagerie Intégré* (SMI), these systems are primarily designed for transaction processing and real-time tracking rather than predictive analytics. As a result, historical data is fragmented across multiple interfaces and is not directly exploitable for statistical modeling or forecasting purposes.

This situation creates a gap between operational data availability and analytical requirements needed for strategic planning.

1.5.1. Operational Need for Forecasting

Outgoing international mail flows are subject to strong variability driven by several factors, including:

- Seasonal effects (e.g., Ramadan-related peaks);
- E-commerce demand fluctuations;
- International logistics constraints;
- Customs delays and regulatory differences;
- Behavioral patterns of the Moroccan diaspora.

Without forecasting tools, operational planning relies heavily on historical averages or expert intuition. This limits the ability to anticipate demand peaks and optimize resource allocation in advance.

Accurate forecasting is therefore essential for:

- Transport capacity planning;
- Sorting center workload management;
- Workforce allocation;
- Service quality improvement;
- Cost optimization in international logistics operations [14].

1.5.2. Data-Related Constraints

A major challenge identified during this internship is the lack of a unified analytical data infrastructure.

The main constraints include:

- Absence of a dedicated data warehouse for historical postal data;
- Lack of structured APIs for bulk data extraction;
- Reliance on web-based reporting interfaces;
- Performance limitations of legacy systems for large queries;
- Incomplete or inconsistent historical coverage across modules.

As a consequence, building a usable dataset requires extensive data engineering efforts, including:

- Web automation for extraction;
- HTML parsing of tracking information;
- Data cleaning and deduplication;
- Transformation into structured formats suitable for analysis.

These constraints significantly increase the complexity of any forecasting pipeline built on top of operational systems.

1.5.3. Analytical Gap

Although BAM systems provide detailed operational tracking, they do not natively support predictive analytics.

The key analytical gap can be summarized as follows:

- Systems provide **descriptive information** (what happened);
- They lack **predictive capabilities** (what will happen);
- Decision-making is therefore reactive rather than proactive.

This gap limits the ability to anticipate demand fluctuations and optimize international logistics flows in advance.

From a data science perspective, this translates into the need for:

- Time-series reconstruction from fragmented sources;

- Robust handling of missing or inconsistent data;
- Selection of appropriate forecasting models adapted to operational constraints.

1.5.4. Problem Formulation

The problem addressed in this internship can be formally defined as a time-series forecasting problem.

Given historical observations of outgoing international mail flows, the objective is to predict future values over a fixed horizon.

Let:

- y_t represent the number of outgoing international parcels (or total weight) at time t ,
- h represent the forecasting horizon.

The goal is to estimate:

$$\hat{y}_{t+h} = f(y_t, y_{t-1}, \dots, y_{t-n}) \quad (1)$$

where f is a forecasting function learned from historical data.

The forecasting task is constrained by:

- Noisy and incomplete historical data;
- Strong seasonal patterns;
- Structural changes over time;
- Limited reliability of long-term extrapolation.

The chosen forecasting horizon is four weeks, which corresponds to operational planning cycles within postal logistics.

1.5.5. Objectives of the Study

The main objectives of this study are:

- Reconstruct a consistent historical dataset from operational systems;
- Identify key temporal patterns in outgoing international mail flows;
- Develop and compare multiple forecasting models;
- Evaluate forecasting performance using robust statistical metrics;
- Provide actionable insights for operational planning.

These objectives aim to transform raw operational data into a structured analytical framework capable of supporting decision-making.

1.5.6. Expected Outcome

The expected outcome of this internship is a complete forecasting pipeline capable of:

- Processing raw operational postal data;
- Generating reliable short-term forecasts;
- Quantifying uncertainty through prediction intervals;
- Supporting planning decisions within Barid Al-Maghrib.

This pipeline is intended to improve visibility over international mail demand and contribute to more efficient resource allocation.

1.5.7. Summary

The core problem addressed in this internship lies at the intersection of operational constraints and analytical requirements.

While Barid Al-Maghrib possesses rich operational data, the lack of structured analytical infrastructure prevents its direct use for forecasting purposes. This internship addresses this gap by proposing a complete data engineering and forecasting pipeline capable of transforming raw operational data into actionable predictions.

2. Methodology

2.1. General Methodological Approach

The methodological framework adopted in this internship is designed to address the challenges associated with extracting, structuring, and forecasting operational data from legacy postal information systems. Given the absence of a dedicated analytical infrastructure, the approach combines data engineering techniques, web automation, and statistical forecasting methods into a unified pipeline.

The overall methodology follows a structured data-driven process composed of four main stages:

- Data acquisition from heterogeneous operational sources;
- Data preprocessing and transformation;
- Time series modeling and forecasting;
- Model evaluation and validation.

This structured approach ensures traceability, reproducibility, and consistency across all stages of the analysis.

2.1.1. Objectives of the Methodological Framework

The primary objective of the methodology is to transform raw operational data into actionable forecasts that can support decision-making in the context of international mail logistics.

More specifically, the methodological framework aims to:

- Extract reliable historical data from operational systems (SMI and tracking interfaces);
- Clean and standardize heterogeneous datasets;
- Construct coherent time series representing postal activity;
- Apply and compare multiple forecasting models;
- Evaluate predictive performance using robust statistical metrics.

This aligns with standard data science pipelines for operational forecasting, where raw transactional data is progressively transformed into structured analytical inputs [15].

2.1.2. Overall Pipeline Overview

The methodological pipeline adopted in this study can be conceptually described as a sequential transformation process:

Raw Operational Systems → Data Extraction → Data Cleaning → Feature Engineering → Forecasting Models → Evaluation

Each stage plays a distinct role:

- Data Extraction: retrieves raw information from SMI reports, tracking systems, and web interfaces;
- Data Cleaning: removes inconsistencies, duplicates, and invalid records;
- Feature Engineering: constructs aggregated time series and relevant explanatory variables;
- Forecasting Models: generate predictions using statistical and machine learning approaches;
- Evaluation: measures predictive accuracy using rolling-origin validation.

This pipeline ensures that each transformation step is explicitly controlled and reproducible, which is essential when working with legacy and heterogeneous systems [16].

2.1.3. Design Principles

The methodological design is guided by three main principles:

2.1.3.1. Robustness

Given the instability of legacy systems and web-based interfaces, the data extraction process must be resilient to interruptions such as session expiration, network failures, and server timeouts. Robust retry mechanisms and checkpointing strategies are therefore integrated into the pipeline.

2.1.3.2. Reproducibility

All data processing steps are designed to be reproducible. This includes:

- Deterministic extraction procedures;
- Versioned intermediate datasets;
- Structured storage using columnar formats (Parquet);
- Script-based automation of all transformations.

Reproducibility ensures that results can be independently verified and updated when new data becomes available.

2.1.3.3. Scalability

The volume of data involved (hundreds of thousands of parcels and millions of operational events) requires scalable processing techniques. To address this, the pipeline relies on:

- Batch processing strategies;
- Efficient data formats (Parquet compression);
- Chunked data extraction to avoid system overload;
- Vectorized operations during preprocessing.

These design principles are consistent with best practices in industrial data engineering systems [17].

2.1.4. Constraints Imposed by Legacy Systems

The methodological approach is strongly influenced by the constraints of Barid Al-Maghrib's legacy information systems.

The main constraints include:

- Lack of a formal API for structured data access;
- Reliance on HTML-based reporting interfaces;
- Limited performance of long-range queries;
- Absence of a centralized analytical data warehouse;
- Fragmentation of data across multiple operational modules.

These constraints make traditional data extraction methods (such as direct database queries or API consumption) infeasible.

As a result, alternative approaches such as Robotic Process Automation (RPA) and browser automation are required to access and reconstruct historical datasets.

This situation is common in legacy logistics and public-sector information systems, where operational continuity is prioritized over analytical accessibility [18].

Consequently, the methodological framework is explicitly designed to operate under partial observability and imperfect data conditions, ensuring that forecasting remains feasible despite system limitations.

2.2. Data Collection Strategy

The data collection strategy constitutes a central component of the methodological pipeline. It defines how raw operational data is extracted from heterogeneous information systems within Barid Al-Maghrib and transformed into structured datasets suitable for analysis.

Given the absence of a unified analytical data warehouse, data must be retrieved from multiple operational systems, each with its own interface, structure, and limitations. The main sources include the *Système de Messagerie Intégré* (SMI), the tracking system, and operational reporting modules.

2.2.1. Data Sources Overview

The data used in this study originates from three main categories of systems:

2.2.1.1. *Système de Messagerie Intégré* (SMI)

SMI is the core operational system responsible for managing postal and parcel activities. It contains structured information related to:

- Shipment deposits;
- Origin and destination data;
- Weight and dimensional attributes;
- Service types;
- Operational status updates.

However, SMI is primarily designed for transactional processing and manual reporting rather than bulk extraction or analytics. As a result, it does not expose a public API or direct database access for analytical purposes [19].

2.2.1.2. Tracking System

The tracking system provides event-level visibility for individual shipments. Each parcel is associated with a sequence of timestamped events representing its lifecycle.

Unlike SMI, the tracking system:

- Requires individual queries per shipment;
- Returns data in HTML format;
- Is optimized for end-user consultation rather than batch extraction;
- Does not support structured export formats.

This makes large-scale extraction computationally expensive and necessitates automation strategies.

2.2.1.3. Reporting Interfaces

Operational reporting modules provide aggregated views of postal activity, including:

- Daily volume reports;
- Agency-level statistics;
- Destination breakdowns;
- Delivery performance indicators.

Although useful for operational monitoring, these reports suffer from:

- Performance degradation for long time ranges;
- Limited export capabilities;
- Lack of structured APIs;
- Inconsistent data granularity.

2.2.2. Absence of Analytical Infrastructure

A key constraint in the data collection process is the absence of a dedicated analytical infrastructure.

Unlike modern data ecosystems that rely on centralized data warehouses, BAM's systems are primarily oriented toward operational execution. Consequently:

- Data is distributed across multiple systems;
- Historical data is not centralized;
- No unified query layer exists;
- Access is limited to graphical interfaces.

This fragmentation significantly increases the complexity of data extraction and integration [17].

2.2.3. Data Extraction Strategy

To overcome system limitations, a multi-layer extraction strategy was implemented.

The approach combines:

- Web automation using Playwright for structured reports;
- Sequential scraping of tracking HTML pages;
- Iterative extraction of time-bounded data windows;
- Local persistence of intermediate results.

This strategy ensures:

- Fault tolerance in case of system failure;
- Resumability of long extraction tasks;
- Minimization of load on operational systems;
- Incremental dataset construction.

The extraction process is designed to be idempotent, ensuring that repeated executions do not duplicate previously collected data.

2.2.4. Chunked Temporal Extraction

Due to performance limitations in SMI reporting tools, large time ranges cannot be queried reliably. To address this, data extraction is performed using a chunked temporal strategy.

Instead of requesting full historical ranges, the system:

- Splits the timeline into daily or weekly intervals;
- Extracts data incrementally per interval;
- Aggregates results into a unified dataset.

This approach significantly improves stability and reduces the risk of server-side timeouts.

2.2.5. Data Collection Pipeline Architecture

The overall architecture of the data collection pipeline is illustrated below.

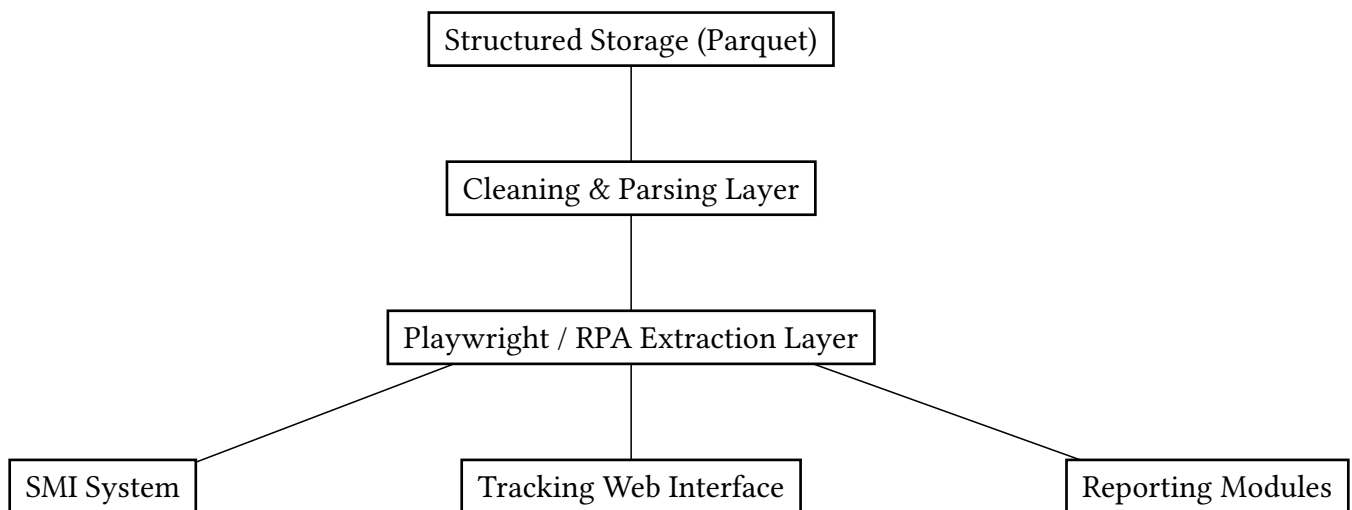


Figure 1: End-to-end data collection pipeline

This architecture highlights the role of the extraction layer as a unifying interface between heterogeneous operational systems and analytical storage.

2.2.6. Data Quality Constraints

Several data quality constraints were encountered during the extraction process:

- Missing values in dimensional attributes;
- Inconsistent status encoding across systems;
- Duplicate records due to repeated exports;
- HTML structure variability in tracking pages;
- Partial or incomplete historical coverage.

To address these issues, multiple validation steps were integrated into the pipeline, including deduplication, schema enforcement, and consistency checks.

2.2.7. Summary

The data collection strategy relies on the integration of multiple heterogeneous systems through automated extraction techniques. Due to the absence of a centralized analytical infrastructure, a custom pipeline based on web automation and incremental extraction was necessary.

This strategy ensures the construction of a consistent and reliable dataset despite operational constraints, forming the foundation for subsequent preprocessing and forecasting tasks.

2.3. Web Scraping and Automation Pipeline

The absence of a formal application programming interface (API) for data extraction from Barid Al-Maghrib's operational systems necessitated the design of a custom web automation pipeline. This pipeline is responsible for extracting structured data from web-based interfaces that were originally designed for human interaction rather than machine-to-machine communication.

The chosen approach is based on Robotic Process Automation (RPA), implemented using the Playwright framework for Python. This enables the simulation of user interactions such as navigation, form submission, and data extraction from dynamic web pages.

2.3.1. Choice of Playwright Framework

Playwright was selected as the primary automation tool due to its ability to:

- Handle modern dynamic web applications;
- Support multiple browser engines (Chromium, Firefox, WebKit);
- Manage persistent sessions and authentication states;
- Interact with JavaScript-rendered content;
- Provide robust selectors for DOM manipulation.

Compared to traditional scraping tools such as BeautifulSoup or Scrapy, Playwright is better suited for applications involving authenticated sessions and dynamically generated content [12].

This makes it particularly appropriate for interacting with SMI and tracking interfaces, which rely heavily on client-side rendering and session-based authentication.

2.3.2. Authentication and Session Management

Access to BAM operational systems requires authenticated sessions. The automation pipeline includes a dedicated login module that:

- Initiates a browser context;
- Submits authentication credentials;
- Stores session cookies;
- Maintains session persistence across multiple requests.

To reduce authentication overhead, session states are cached locally and reused whenever possible. When a session expires, the system automatically re-authenticates.

This mechanism ensures continuity of long-running extraction tasks without manual intervention.

2.3.3. Navigation and Form Automation

Data extraction from SMI and reporting interfaces involves navigating complex web forms with multiple filters such as:

- Date ranges;
- Destination countries;
- Service types;
- Agency identifiers.

The automation pipeline simulates user interaction by:

- Filling input fields programmatically;
- Selecting dropdown values;
- Triggering report generation buttons;
- Waiting for asynchronous page loads;
- Extracting rendered HTML tables.

This process allows structured reports to be generated programmatically despite the absence of API endpoints.

2.3.4. Robustness and Fault Tolerance

Given the instability of legacy systems under heavy queries, robustness is a critical requirement of the pipeline.

A multi-layer fault-tolerant architecture was implemented, including:

2.3.4.1. Retry Mechanism

Each critical operation is wrapped in a retry strategy that attempts to recover from transient failures such as:

- Network timeouts;
- Server-side errors;
- Page load failures;
- Session expiration.

If an operation fails, the system retries after a short delay, up to a predefined maximum number of attempts.

2.3.4.2. Session Recovery

In case of persistent failure, the pipeline:

- Terminates the current browser context;
- Re-initializes a new session;
- Re-authenticates automatically;
- Resumes execution from the last successful checkpoint.

This ensures that long extraction processes can run continuously without manual supervision.

2.3.4.3. Checkpointing Strategy

Progress is stored incrementally to avoid data loss. Each successfully extracted batch is saved locally, allowing the system to resume exactly where it stopped in case of interruption.

2.3.5. Idempotent Extraction Design

To ensure data consistency, the extraction process is designed to be idempotent.

Before processing a given data chunk, the system:

- Checks whether the corresponding data already exists locally;
- Skips extraction if the dataset is already complete;
- Avoids duplication of previously processed records.

This is particularly important when dealing with large-scale historical datasets spanning hundreds of thousands of shipments.

2.3.6. Chunked Extraction Strategy

Due to performance limitations in SMI reporting tools, large-scale queries often fail or become unresponsive. To mitigate this issue, data extraction is performed using a chunked approach.

The timeline is divided into small intervals (typically daily), and each interval is processed independently.

This strategy ensures:

- Reduced server load;
- Lower probability of timeout errors;
- Improved stability of extraction jobs;
- Easier recovery from partial failures.

2.3.7. HTML Parsing and Data Extraction

Once web pages are successfully retrieved, the raw HTML content is parsed to extract structured information.

The extraction process includes:

- Identification of HTML table structures;
- Extraction of tabular rows and columns;
- Normalization of field names;
- Conversion into structured dataframes.

Libraries such as lxml are used to efficiently parse large HTML documents and extract relevant fields.

2.3.8. Automation Pipeline Architecture

The complete automation pipeline is illustrated below.

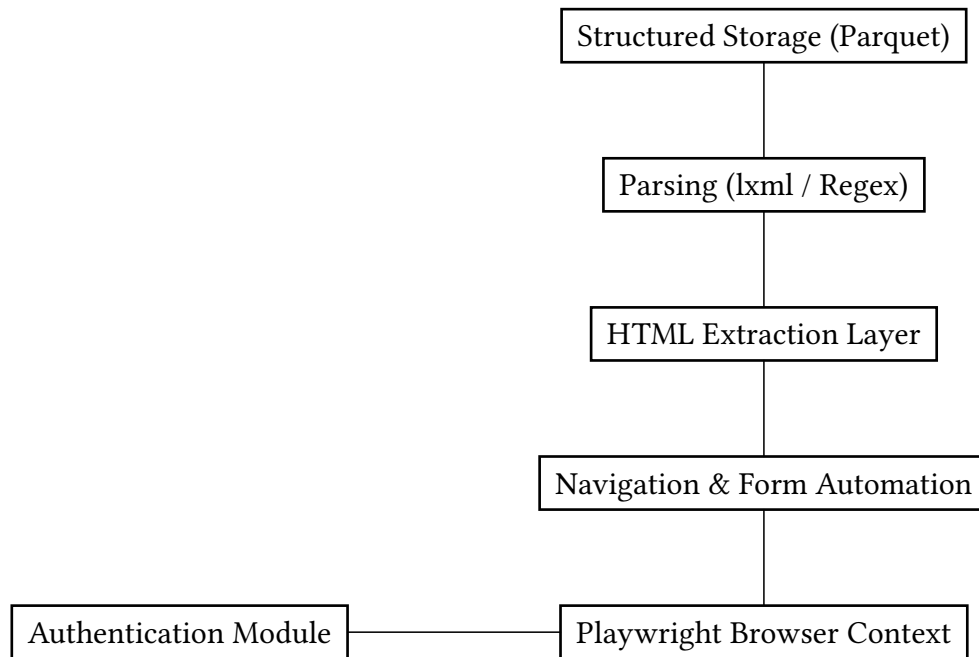


Figure 2: Playwright-based web scraping and automation pipeline

This architecture highlights the sequential transformation from authenticated browser sessions to structured analytical datasets.

2.3.9. Limitations of Web Automation

Despite its effectiveness, the web automation approach presents several limitations:

- High sensitivity to changes in web interface structure;
- Increased execution time compared to direct API access;
- Dependency on browser stability;
- Potential blocking or throttling by backend systems;
- Maintenance overhead due to UI updates.

These limitations are inherent to scraping-based approaches and reinforce the need for more modern API-driven architectures in operational systems.

2.3.10. Summary

The web scraping and automation pipeline provides a robust solution for extracting structured data from legacy operational systems lacking API access. By combining Playwright-based automation, retry mechanisms, and chunked extraction strategies, the system ensures reliable data collection under unstable and constrained conditions.

This pipeline forms a critical component of the overall data engineering architecture used in this internship.

2.4. 2.4 Data Engineering Pipeline (ETL)

The data engineering pipeline constitutes the transformation layer between raw extracted operational data and structured datasets used for analytical modeling. In the context of this internship, the pipeline is responsible for integrating heterogeneous sources, cleaning inconsistent records, and producing time-consistent datasets suitable for forecasting.

Given the complexity and fragmentation of the source systems (SMI, tracking interfaces, and reporting modules), a robust Extract–Transform–Load (ETL) architecture was implemented to ensure data reliability, scalability, and reproducibility.

2.4.1. 2.4.1 ETL Architecture Overview

The ETL pipeline is composed of three main stages:

- Extract: retrieval of raw data from web interfaces and exported files;
- Transform: cleaning, normalization, and feature engineering;
- Load: storage into optimized analytical formats.

This structure ensures a clear separation between raw data acquisition and analytical processing, improving maintainability and traceability [20].

2.4.2. 2.4.2 Extract Phase

The extraction phase has been detailed in previous sections (SMI, tracking, and reporting systems). In summary, it includes:

- Web automation via Playwright;
- Chunked temporal extraction (daily or weekly windows);
- HTML parsing of tracking events;
- Export of structured reports from SMI interfaces.

All extracted data is stored in raw form to ensure traceability and reproducibility.

The guiding principle of this phase is to preserve the original structure of the data as much as possible before transformation.

2.4.3. 2.4.3 Transform Phase

The transformation phase is the most critical component of the pipeline. It involves converting raw, heterogeneous, and often inconsistent data into structured analytical datasets.

2.4.3.1. Data Cleaning

The following cleaning operations were applied:

- Removal of duplicate records using unique identifiers (e.g., `codeenvoi_`, `cab`);
- Filtering of invalid or corrupted status values;
- Standardization of categorical variables (status codes, service types);
- Handling of missing values in numerical attributes;
- Filtering of inconsistent historical records.

2.4.3.2. Schema Standardization

Due to differences between systems, a unified schema was defined. This includes:

- Standardized date formats (ISO 8601);
- Consistent naming conventions for variables;
- Harmonized identifiers across systems;
- Unified representation of shipment status codes.

2.4.3.3. Feature Engineering

Additional variables were constructed to support time series analysis:

- Weekly aggregation of parcel counts;
- Total shipped weight per time unit;
- Average parcel weight;
- Seasonal indicators (e.g., Ramadan period flags);
- Log-transformed variables for variance stabilization.

These transformations improve the suitability of the dataset for statistical modeling and forecasting methods.

2.4.3.4. Temporal Aggregation

Since forecasting is performed at a weekly level, raw transactional data was aggregated using:

- Grouping by week of deposit;
- Summation of counts and weights;
- Computation of summary statistics.

This step ensures consistency between operational granularity and forecasting resolution.

2.4.4. 2.4.4 Load Phase

The final stage of the pipeline consists of storing transformed datasets into efficient analytical formats.

The chosen format is Apache Parquet due to its advantages:

- Columnar storage optimized for analytical queries;
- High compression ratios;
- Efficient I/O performance;
- Compatibility with Python data science ecosystem.

Storing intermediate and final datasets in Parquet format ensures:

- Fast loading during model training;
- Reduced storage footprint;
- Long-term reproducibility of results.

2.4.5. 2.4.5 Pipeline Automation and Orchestration

The ETL pipeline is fully automated using Python scripts orchestrating sequential tasks:

- Scheduled execution of extraction modules;
- Batch processing of transformation steps;
- Automatic validation of outputs;
- Logging of pipeline execution stages.

A modular design is used to ensure that each component can be executed independently or as part of a full pipeline run.

2.4.6. 2.4.6 Data Validation and Quality Control

To ensure data integrity, several validation rules were implemented:

- Referential integrity checks across datasets;
- Range validation for numerical variables (e.g., weight > 0);
- Consistency checks between tracking and deposit data;
- Detection of missing temporal periods;
- Outlier detection using statistical thresholds.

Records failing validation are either corrected when possible or excluded from the final dataset.

2.4.7. 2.4.7 Storage Optimization

Given the large volume of data (millions of operational records), storage optimization techniques were applied:

- Columnar compression using Parquet;
- Removal of redundant raw HTML after parsing;
- Deduplication of intermediate datasets;
- Partitioning by year and dataset type.

These optimizations significantly reduce storage requirements while maintaining analytical fidelity.

2.4.8. 2.4.8 ETL Pipeline Architecture

The overall ETL architecture is illustrated below.

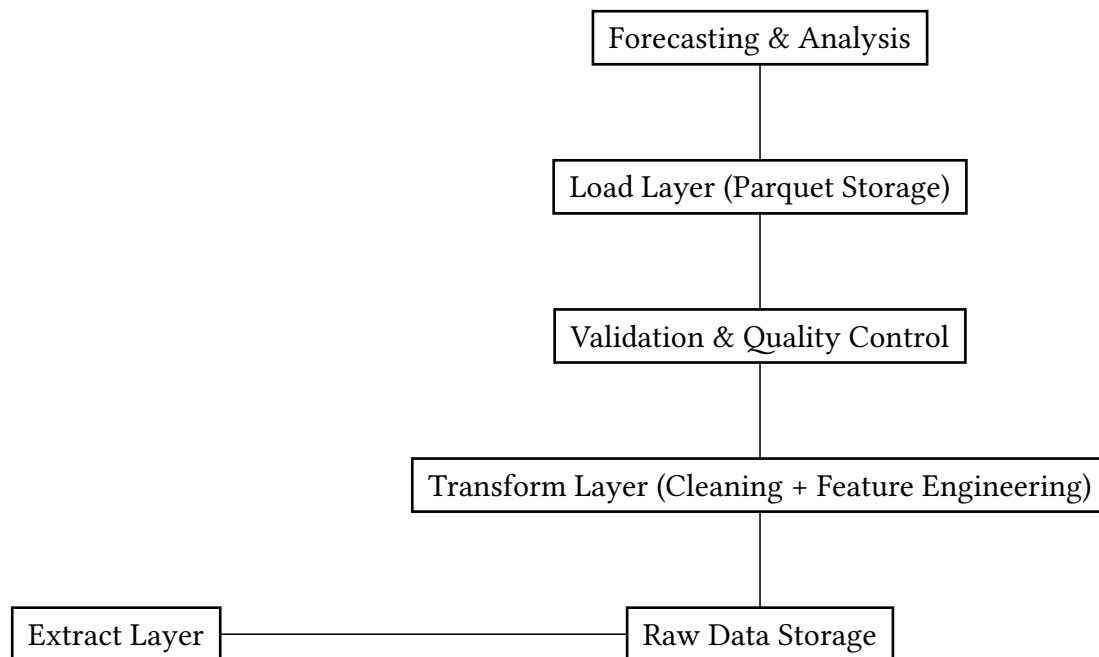


Figure 3: End-to-end ETL pipeline for postal data processing

2.4.9. 2.4.9 Challenges in Data Engineering

Several challenges were encountered during ETL implementation:

- Heterogeneity of source systems;
- Inconsistent historical records across platforms;
- Large-scale data volume (millions of records);
- Performance limitations during transformation;
- Dependence on previously extracted raw data integrity.

These challenges required iterative refinement of both extraction and transformation logic.

2.4.10. 2.4.10 Summary

The ETL pipeline provides a structured framework for transforming raw operational data into clean, consistent, and analysis-ready datasets. By combining automated extraction, robust transformation logic, and optimized storage formats, the system ensures that downstream forecasting models operate on high-quality data.

This pipeline represents the core data engineering contribution of this internship project.

2.5. Construction of the Forecasting Dataset

The objective of the forecasting stage is to predict future international postal activity based on historical observations. However, the data collected from operational information systems are recorded at the shipment level and therefore cannot be directly used by forecasting models. A dedicated processing stage was consequently implemented to transform transactional data into regularly spaced time series suitable for statistical analysis.

2.5.1. Selection of the Reference Dataset

Several datasets were collected during the extraction phase, including deposit records, parcel tracking events, delivery information and optional service data. Although these datasets provide complementary perspectives on postal operations, not all of them are suitable for demand forecasting.

The **Envois by produit international** dataset was selected as the primary forecasting source because it contains the date of deposit for every outgoing international shipment and exhibits the highest level of completeness over the study period.

Tracking information was not used directly for forecasting purposes. Once parcels leave Morocco, subsequent events are often unavailable or only partially observed. Consequently, the deposit date constitutes the earliest and most reliable indicator of outgoing international demand.

Delivery-related datasets and optional service information were retained for exploratory analyses but were excluded from model training.

2.5.2. Temporal Scope of the Study

Preliminary investigations revealed that historical records prior to 2023 are sparse and highly irregular. Large periods contained very few observations, making them unsuitable for robust estimation of seasonal components.

To ensure sufficient temporal consistency, only observations collected between January 2023 and April 2026 were retained.

This filtering step improves the reliability of statistical inference while maintaining an observation window sufficiently long to capture annual seasonal patterns.

2.5.3. Definition of Forecast Variables

The forecasting task focuses on two operational variables:

- Number of deposited parcels;
- Total deposited weight.

These variables were selected because they directly influence several operational decisions, including:

- transportation planning;
- workforce allocation;
- international dispatch scheduling;
- storage capacity management.

Parcel dimensions were intentionally excluded from the forecasting process due to their high proportion of missing values, particularly before 2026.

2.5.4. Weekly Aggregation

Raw shipment data are available at the individual parcel level. Since postal demand exhibits strong day-to-day fluctuations and operational planning is generally performed over weekly horizons, observations were aggregated by calendar week.

For each week t , the following quantities were computed:

$$\text{Count}_t = \sum_i 1$$

and

$$\text{Weight}_t = \sum_i w_i$$

where w_i denotes the real weight associated with shipment i .

Weekly aggregation presents several advantages:

- reduction of daily noise;
- mitigation of holiday effects;
- improved stability of seasonal estimates;
- compatibility with medium-term planning horizons.

2.5.5. Seasonality Investigation

Exploratory analyses revealed a recurrent increase in parcel volumes during the weeks preceding Ramadan.

This phenomenon can be explained by the significant Moroccan diaspora residing abroad, which frequently sends food products, gifts and personal items to relatives before the beginning of Ramadan.

Visual inspection suggests that this effect is stronger than conventional annual seasonality and therefore represents an important characteristic of the studied series.

No explicit Ramadan regressors were introduced in this study. Instead, forecasting models were expected to capture these patterns implicitly through historical observations.

2.5.6. Variance Stabilization

Forecasting methods such as ETS and ARIMA often assume approximately constant variance over time.

Visual analysis indicated that fluctuations tend to increase during periods of high postal activity.

A logarithmic transformation was therefore applied:

$$z_t = \log(y_t)$$

where y_t represents either weekly parcel counts or weekly deposited weight.

The transformation offers several advantages:

- reduction of heteroscedasticity;
- improved numerical stability;
- guaranteed positivity after inverse transformation.

Forecasts generated on the transformed scale were converted back to their original units through the exponential function.

2.5.7. Forecasting Horizon

The forecasting problem was formulated as a four-week-ahead prediction task.

A forecasting horizon of four weeks was selected because it provides a reasonable compromise between operational usefulness and predictive reliability.

Shorter horizons generally offer limited planning capabilities, whereas longer horizons tend to accumulate uncertainty.

The objective can therefore be written as:

$$\hat{y}_{t+h}$$

for

$$h \in \{1, 2, 3, 4\}$$

where each forecast corresponds to one future week.

2.5.8. Final Dataset Structure

After preprocessing and aggregation, the final analytical dataset contains:

- weekly timestamp;
- parcel count;
- total deposited weight;
- mean parcel weight;
- transformed variables.

The resulting series constitute the input used by all forecasting models evaluated in the following chapter.

2.5.9. Summary

The construction of the forecasting dataset transforms heterogeneous shipment-level observations into coherent weekly time series representative of Morocco's outgoing international postal activity.

This stage bridges the gap between operational information systems and statistical forecasting models, ensuring that the subsequent analysis is performed on stable, interpretable and operationally meaningful indicators.

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